

GramAI: SIH Architecture Blueprint

An AI Advisor & Smart Loan Router Architecture engineered for the Smart India Hackathon (SIH)

Problem Statement Focus: Enabling rural entrepreneurs (farmers, tailors, shopkeepers) to successfully leverage government SCA loans (10/90 rule) by bridging the gaps in local language understanding, hyper-local market feasibility analysis, and frictionless voice-driven daily bookkeeping.

1. Core System Architecture Layering

To satisfy strict validation benchmarks by SIH judges, the pipeline balances low bandwidth consumption, zero-hallucination factual query mapping, and offline resilience:

A. Mobile Client Application Layer (Flutter / React Native)

- **Offline State Sync:** Implements SQLite local storage to act as the primary voice ledger buffer.
- **Edge-VAD Execution:** Hosts Silero Voice Activity Detection directly on-device to cleanly slice audio payloads, reducing payload sizes over fragile 2G/3G networks.

B. The Vani-Voice Processing Pipeline (Speech-to-Intent)

- **ASR Framework:** Direct mapping into India's Bhashini Infrastructure API or an orchestration layer driving multi-lingual Whisper models optimized for vernacular dialects (Hindi, Marathi, Kannada).
- **Deterministic Entities Extraction:** Employs fine-tuned regex matching patterns alongside highly-structured JSON formatting layers to cleanly separate variables (e.g., location, savings amount, chosen trade).

C. The Gram-RAG Intelligence Core

- **Deterministic State Routing:** Rather than handling raw financial operations via an LLM, loan calculations are run programmatically through standard logic loops inside FastAPI matching State Channelizing Agency guidelines.
- **Knowledge Management:** Feeds translated context into a Vector DB (Qdrant or Milvus) loaded with localized regional farming frameworks, disease data, and operational guidelines.

2. Proposed Technical Specification Matrix

Infrastructure Layer	Recommended Component Stack	Strategic Advantage for SIH Evaluation
Frontend Mobile App	Flutter or React Native + local SQLite store	Provides instantaneous response feedback and native audio drivers; seamlessly buffers transactions locally.
Backend API Engine	FastAPI (Python) + WebSockets	Asynchronous execution handles thousands of concurrency vectors with rapid processing speeds for multi-modal assets.
Voice Pipeline	Bhashini Government API Core + Silero VAD	Direct alignments with National Digital Public Infrastructure architectures, gaining immediate compliance appreciation.
Orchestration & RAG	LlamaIndex Framework + Qdrant / Milvus Vector DB	Strict control parameters over external retrieval processes, ensuring rapid embedding lookups.
Primary Database & GIS	PostgreSQL + PostGIS Extension Data Fabric	Powers high-velocity geometric calculations for complex hyper-local tracking (5-10km trade clusters).

3. X-Factor Project Integrations for a Winning Edge

Implementing generic AI systems will not suffice for the top rank. Incorporating these localized production modifications proves true industry preparedness:

1. Detailed Project Report (DPR) Auto-Generation

Instead of presenting raw unstructured visual cards, compile extracted data vectors into an iron-clad business proposal PDF matching standard institutional formatting. Dynamically run calculations to show accurate Cash Flow projections, Debt Service Coverage Ratios (DSCR), and Internal Rate of Return (IRR) metrics to thoroughly impress industrial banking judges.

2. Alternative Credit Profile Scoring (Micro-Lending Risk Engine)

Given that typical rural micro-entrepreneurs lack a formal CIBIL score tracking baseline, leverage the continuous logs from the *Bol ke Khata* system. Build a custom metrics scoring array mapping revenue stability, low inventory drift, and cash reliability to generate alternative proof-of-credit data structures exportable to local banks.

3. WhatsApp & Audio Telephony IVR Fallback

Expose an auxiliary ingestion endpoint using a basic IVR toll-free channel or structured WhatsApp bot loops. If an end-user lacks smartphone access, they can place a standard phone call, register transaction data, and listen to automatically configured financial status statements over text-to-speech feedback pipelines.

This blueprint is created for hackathon ideation and software planning purposes only. AI systems require rigorous edge-case handling before production deployment.